

Developments in diagnostic techniques for acute myocardial infarction (heart attack) and the implications for Critical Illness claims

Overview and aims of the paper

Whilst the majority of claims for heart attack are relatively straight forward and do not present any problems at claims assessment, there have been significant developments in recent years in the diagnosis of heart attack or myocardial infarction (MI) that are impacting upon claims and it is vitally important that claims assessors are aware of the medical developments in this area.

This paper explores how these changes have impacted upon the common understanding of the term 'heart attack' and provides information with the intention of assisting claims assessors in working through the complexities involved. In addition, we hope that this information will also help companies in keeping their claims philosophies contemporary with developments and to help ensure that all claims within the industry are dealt with in a fair and consistent manner.

There have been significant developments in recent years in the diagnosis of MI. These include the worldwide acceptance of revised diagnostic criteria for MI in 2007 and more recently significant developments in the use of increasingly sensitive "troponins", blood tests that are used in diagnosing MI.

Troponins are types of biochemical cardiac markers that are released into the blood following MI. The specific types tested for are troponin I or troponin T and have been used widely in the diagnosis of MI for over 10 years. More recently there has been the development of "high sensitivity" troponins that have made it possible to diagnose MI in cases where historically it would not have been possible to do so.

In producing this paper, the working group has consulted widely with Chief Medical Officers within the industry and with Dr Paul Collinson¹, a leading expert in the fields of cardiology/biochemistry and the use of troponins.

Diagnosis of MI and the use of Troponins

The underlying process that leads to MI is usually ischaemic heart disease. This is a condition where the coronary arteries that supply blood and oxygen to the heart muscle, become narrowed because of a build-up of a substance called atheroma. If the blood flow is blocked, the heart is starved of oxygen and heart cells die. Plaques of atheroma may gradually form over a number of years in one or more places in the coronary arteries. Each plaque has an outer firm shell with an inner core.

With MI what usually happens is that a crack develops in the outer shell of the atheroma (plaque rupture) or the surface of the plaque erodes (plaque erosion). This exposes the inner core of the plaque to blood, that in turn can trigger the clotting mechanism in the blood to form a blood clot within the coronary artery affected. With MI the blood supply becomes completely obstructed, which leads to the sudden death of a portion of the heart muscle. Less commonly, MI may be caused by the slow build-up of plaque which gradually blocks the patency of a coronary artery.

However, not all those suffering ischaemic heart disease experience MI. If all cases of known ischaemic heart disease were covered, critical illness cover would be substantially more expensive and, for many people, unaffordable.

The diagnosis of MI is not always easy. Electrocardiograms (ECGs) are typically used to help diagnose MI. However, the ECG test alone is not diagnostic and traditionally doctors would also use a blood test to measure biochemical markers released into the blood when cells in the heart muscle die. The markers used, historically were known as “cardiac enzymes”, which have now almost totally been replaced over time by the more sensitive and accurate “troponins”.

The increased sensitivity of troponins for detecting heart muscle cell damage means they have a prognostic value in identifying patients who are at an increased risk of subsequent serious episodes of ischaemia or MI, or who will require a coronary angioplasty or surgical by-pass grafting. Therefore, these tests assist in the risk classification of patients and in arranging the most appropriate care or preventative measures.

That being said, there are times when troponin may not be recorded in cases of suspected MI. This is often the case where the ECG at a time of an acute episode indicates definite changes of myocardial ischaemia with persistent ST segment elevation, when the attending physician may consider there to be no value in obtaining biochemical markers. However, practices vary in this approach and others may still choose to record troponin in view of the relatively small expense involved and value it still might provide from a diagnostic perspective.

The specific types tested for are troponin I or troponin T, which have been used widely in the diagnosis of MI for over 10 years. More recently there has been the development of “high sensitivity” troponins that can detect minute cellular damage which make it possible to diagnose MI in cases where historically some cases would have been categorised as suffering from “acute coronary syndrome” without MI.

Troponins naturally degrade and following MI there is a typical pattern of a rise and fall in the levels present. However, as with cardiac enzymes, the level of troponin may be raised for reasons other than MI. For example, this can happen after surgical procedures or in the presence of myocarditis. This means that there will be patients who have raised troponin levels but who have not suffered a heart attack. This area is expanded upon a little later.

The current ABI model definition for heart attack offers some future-proofing to ensure there is continuation of claims paid where similar events have occurred over time, as it states “Heart attack of specified severity” and requires within the definition that particular levels of troponin must be exceeded to meet the claims criteria. However, the developments in diagnosing MI are creating some problems as it has been noticed recently that there appears to be an increase in the number of claims being seen where the troponin levels required under the definition are not being met, even though the claimants may be told they have suffered a “heart attack” and are expecting to be covered.

The medical definition of MI

The first universal medical definition for MI was published by the paper “*Universal definition of myocardial infarction*”² produced in 2007 by a task force specifically brought together to redefine myocardial infarction and made up by the following medical committees; the Joint European Society for Cardiology (ESC)/American College of Cardiology Foundation (ACCF)/American Heart Association (AHA)/ World Heart Federation (WHF).

The wording is as follows:

Criteria for acute myocardial infarction

The term myocardial infarction should be used when there is evidence of myocardial necrosis in a clinical setting consistent with myocardial ischaemia. Under these conditions any of the following criteria meets the diagnosis for myocardial infarction:

Detection of rise and/or fall of cardiac biomarkers (preferably troponin) with at least one value above the 99th percentile of the upper reference limit (URL) together with at least evidence of myocardial ischaemia with one of the following:

- Symptoms of ischaemia;
- ECG changes indicative of new ischaemia [new ST-T changes or new left bundle branch block (LBBB)];
- Development of pathological Q waves in the ECG;
- Imaging evidence of new loss of viable myocardium or new regional wall motion abnormality.

The definition then goes on to detail how myocardial infarction is diagnosed when someone has died before tests can be performed or when surgical procedures to the heart, namely coronary angioplasty or bypass have been undertaken and MI is suspected to have occurred at the same time.

The previous MI definition was produced by the World Health Organisation (WHO) in the 1970's and defined MI from symptoms, ECG abnormalities and enzymes released into the blood, rather than troponins.

One of the major differences in the new definition was placing the importance of biochemical markers with the specific mention of troponin at the cornerstone of how a diagnosis of MI is made.

When deciding on medical definitions for critical illness it has always been important as far as possible to use a definition close to that used by the medical profession. Some changes may then be needed so that the wording is appropriate for an insurance policy. However, the revised MI definition is longer and more medically complex than the previous WHO definition. Therefore, it may prove difficult to incorporate some of the features it contains for use within the heart attack definition contained within the Statement of Best Practice, from the perspective of making it too complex for a consumer to be expected to understand.

The mention of a 99th percentile referred to here is worthy of some explanation as it is also referred to later in the paper. In this context, it is the level of troponin at which 99 out of 100 people in a healthy population will have a negative result. i.e. 1 in a 100 will have a false positive result. The manufacturers of the assays define the 99th percentile values for a particular assay which vary according to the sensitivity of each assay.

In August 2012 a third universal definition of myocardial infarction was published which is still to be ratified. It includes additional diagnostic criteria for acute MI on top of those accepted in 2007. The new diagnostic criteria allows for the identification of an intracoronary thrombus on angiography or autopsy as evidence of an acute myocardial infarction. However, this third universal definition does not affect the main constituent parts for MI diagnosis as detailed above.

The ABI model definition for heart attack

The current ABI Statement of Best Practice for Critical Illness Cover was published in 2011 and includes 23 critical illnesses that are typically covered.

These are set as an 'appropriate minimum standard' so that insurers have to use that definition, or may modify the definition in prescribed ways to offer at least the same cover. We should therefore always try to consider potential changes to definitions at an early stage. It is important to note that most insurers cannot normally amend the definitions of critical illnesses for policies that are already in force except, in some circumstances, to improve the cover.

In practice, some companies will extend the cover provided by the ABI definitions, in these circumstances they refer to the definition as "ABI+" within their literature, indicating that they have changed the wording from that provided by the ABI. This is done with the aim of trying to offer better cover for consumers and also for marketing purposes when product features are being compared.

Currently, companies state they are offering ABI+ cover with regards to "heart attack" by removing the part of the wording relating to "typical clinical symptoms" on the basis that some heart attacks may be discovered without an acute episode (silent myocardial infarction).

The current ABI model definition for heart attack is as follows:

Heart attack – of specified severity

Death of heart muscle, due to inadequate blood supply, that has resulted in all of the following evidence of acute myocardial infarction:

- Typical clinical symptoms (for example, characteristic chest pain).
- New characteristic electrocardiographic changes.
- The characteristic rise of cardiac enzymes or Troponins recorded at the following levels or higher;
 - Troponin T > 1.0 ng/ml
 - AccuTnI > 0.5 ng/ml or equivalent threshold with other Troponin I methods.

The evidence must show a definite acute myocardial infarction.

For the above definition, the following are not covered:

- Other acute coronary syndromes including but not limited to angina.

The definition aims to provide cover for all heart attacks of "specified severity" where there is evidence in the form of typical clinical symptoms, new characteristic ECG changes, characteristic rise of enzymes of MI or troponins that meet a stated value.

This has been the ABI's model definition since 2006 and was the first to state a required value for troponins. The decision to do so was made before the revised universal, medical MI definition had been finally agreed and published.

At the time there were mixed views within the medical profession as to whether a diagnosis of MI should be made with lower values of troponin, where historically cardiac enzymes because of their inferior sensitivity would not have confirmed diagnosis. In 2004 the British Cardiac Society (BCS) released a paper³ suggesting that a diagnosis of MI might only be made where traditional cardiac enzymes would have confirmed diagnosis and produced the following table:

Spectrum of acute coronary syndrome		
ACS with unstable angina	ACS with myocyte necrosis	ACS with clinical myocardial infarction
Marker: Tn and CK-MB undetectable	Troponon elevated Tn T < 1.0 ng/ml	Tn T > 1.0 ng/ml +/- CK-MB↑ or AccuTn I > 0.5 ng/ml
ECG: ST↓ or T↓ or transient ST↑ or normal		ST↑ or ST↓ or T inversion: may evolve Q waves
Risk of death*: 5–8%	8–12%	12–15%
Pathology: plaque disruption, intra-coronary thrombus, micro-emboli		
partial coronary occlusion	←→	complete coronary occlusion
LV function:		
no measurable dysfunction	←→	systolic dysfunction, LV dilatation

The paper proposed that the troponin thresholds for defining MI were set at the values stated above, on the basis that the risk of death was similar to that seen using traditional cardiac enzymes and the previous WHO definition. Importantly, these levels were setting a higher limit than the Universal MI definition, which suggested the use of the 99th percentile of the URL (upper reference limit).

The BCS paper was extremely helpful in providing information to keep the critical illness heart attack definition contemporary and enabled cover to be provided for cases at the same historic level of severity, before troponin had started to be used in the diagnosis of MI. This was particularly important at this time, as it was uncertain where the revised medical definition for MI might end up, and provided the actual values of troponin that could be used within the definition itself and applied when assessing claims. Therefore, the ABI definition adopted these values for troponin within its definition and it has worked well until the recent significant developments mentioned in this document.

The BCS paper also had some influence in how MI was reported in the UK up until 2007 when the universal definition was published. From that time on MI incidence has been reported based on the revised universal MI definition.

During the discussions leading up to the production of this paper, a very important consideration was raised by several CMOs and also by Dr Collinson relating to the value used for troponin T that was set back in 2004 by the BSC. The view being that the figure currently stated should be reconsidered in light of more current thinking and the improved knowledge that now exists following the greater use of troponins over time.

Dr Collinson felt a troponin T level of 200 ng/L (0.2 μ g/L or 0.2 ng/ml) could generally be considered an equivalent with the enzyme levels used to diagnose myocardial infarction by the previous WHO definition. Dr Collinson also confirmed that the troponin T level would always be a lower figure than troponin I if they were to be taken simultaneously following myocardial infarction which is at odds with the current ABI definition.

Previous ABI model wordings for Heart Attack

Model wordings for heart attack were issued in the 1999 and 2002 ABI Statements of Best Practice for Critical Illness and can be found in Appendix1.

The 1999 definition only makes reference to “cardiac enzymes” with no reference to troponin, whilst the 2002 wording mentions troponins but does not include a severity level that needs to be met. Therefore, any claims with policies where these wordings apply are unlikely to be affected by the changes in diagnosing heart attack and claims are more likely to be met with any troponin increase providing they are in the context of MI.

The impact of “high sensitivity” troponins

As has been mentioned already, within the last few years there have been significant developments in the production of “high sensitivity” troponins that are able to detect cellular damage to the heart muscle at even lower levels than the traditional troponin assays. Therefore, with the acceptance in the UK of the 2007 universal MI definition with its criteria of troponin being detected over the 99th percentile, this has led to more cases of acute coronary syndrome being diagnosed as MI. Quite how much this has increased is difficult to say, but it is the view of some of the CMOs consulted that it could be as high as 20% or 30%.

The term “high-sensitivity” (hs) reflects the assay’s characteristics and does not refer to a difference in the form of cardiac troponin being measured. It is the same molecules being tested for troponin I or T, just more sensitive assays being used to detect them.

The use of hs troponins in hospitals has spread very rapidly throughout the UK and it is only a matter of time before all hospitals are using these more sensitive types of assay.

There has also been a move within the medical world to change the unit values from ng/ml to ng/L. This is because the sensitivities are leading to much smaller numbers at the 99th percentile becoming detectable. This becomes problematical as the more zeros pass the decimal point the greater the risk where the rounding of decimals to zero can result in clinical errors in data reporting for both electronic medical records and electronic data transfer. This change in practice has resulted in more claims being seen using the different units that also appears to have caused some confusion with claims assessors where this change has not been understood.

There are a number of troponin assays that are being manufactured and used in the UK. For troponin T there is only one manufacturer (Roche) as they have issued a patent on its production. However, there are 20+ assays being manufactured for troponin I by various companies, although not all are available for commercial use or are being used in clinical practice.

The various assays that measure troponins have different sensitivities at the lower end of the scale and so have differing 99th percentile values. However, they are testing for the same molecules i.e. troponin T or I and so, the values found at the higher levels are consistent regardless of the assay used.

Further possible changes

It is generally considered that the use of troponins is unlikely to be replaced in the foreseeable future and this is also the view of Dr Collinson. What this means in simple terms is that for the next ten years at least, the use of troponin is likely to remain at the cornerstone in diagnosing MI.

A recent paper published titled "*Implementation of a Sensitive Troponin I Assay and Risk of Recurrent Myocardial Infarction and Death in Patients With Suspected Acute Coronary Syndrome*"⁴, details the effects of accepting greater assay imprecision and serves as an example of how changes could be made to the universal definition. The universal MI definition currently requires a coefficient of variation (COV) of $\leq 10\%$ in order to minimise the risk of false positive results when evaluating troponin.

The above paper suggests that changing the COV to $\leq 20\%$ would identify more patients at risk of recurrent myocardial infarction, but would increase the diagnosis of MI by 47% by effectively moving those within the category of unstable angina into MI incidence. Dr Collinson's view is that on the basis of this paper alone it is unlikely that the universal MI definition will change. This was confirmed by the recent third universal definition of myocardial infarction.

Issues with the current ABI Heart Attack model wording

One issue with the current model wording relates to the troponin values quoted. Firstly they are quoted using the older unit values. There is evidence to suggest this has definitely caused problems with claims where assessors have been unaware and have not understood this development. Therefore, careful consideration should be taken when considering troponin values and a simple conversion is explained below using the current definition:

The characteristic rise of cardiac enzymes or troponins recorded at the following levels or higher;

- Troponin T > 1.0 ng/ml - equal to 1000 ng/L or 1.0 $\mu\text{g/L}$
- Troponin I > 0.5 ng/ml - equal to 500 ng/L or 0.5 $\mu\text{g/L}$

As already mentioned, there is also the issue concerning the troponin T value itself and it has been suggested this should be lowered to reflect more accurately the previous enzyme equivalent values used in diagnosing MI.

A minor issue concerns reference to part of the definition:

For the above definition, the following are not covered:

- Other acute coronary syndromes including but not limited to angina.

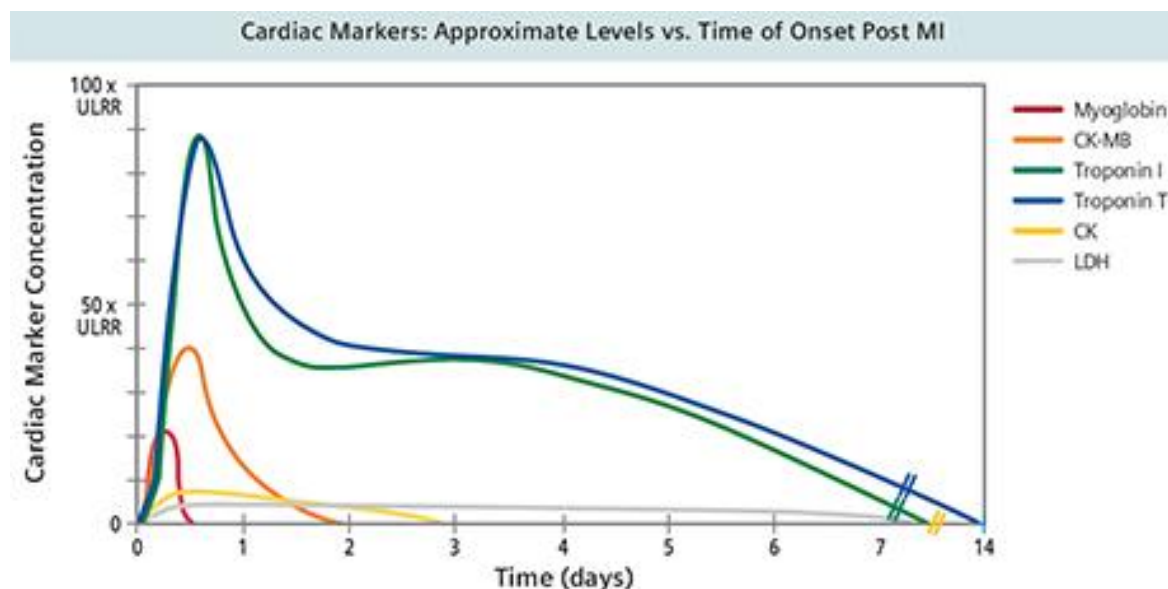
This is because from a medical perspective it is "unstable angina" rather than "angina" that is recognised as part of the spectrum of conditions included within "acute coronary syndromes". Whilst this part of the definition does not create any claims problems, it could result in criticism by questioning its strict technical accuracy.

The existing wording also refers to an assay “AccuTnI” which at the time the definition was drafted was one of the most common assays used for measuring troponin I. However, the assay has been superseded and is no longer in clinical use. Therefore, it would be appropriate to state a level of troponin I that can be applied to any troponin I assay as shown in the example above.

Other important claims issues being encountered

In addition to cases being diagnosed as MI that do not meet the criteria because of the current troponin values in the definition, there are problems due to troponins not regularly being taken when they reach their peak. As mentioned earlier, following MI, troponin is released into the blood and there is a typical pattern of a rise and fall in the levels. So, claims assessors are sometimes trying to estimate what the troponins might have been in order to meet with the value contained in the definition had they been taken at a different time.

To explain this more, the graph below, produced by one of the manufacturers of troponin assays, shows the typical pattern of troponin levels following MI.



This shows that elevation in troponins start within a few hours and peak somewhere between 12 to 24 hours. This information can vary study to study because there is variation in the ranges and durations due to the size of the events. So a very large MI may take longer to reach a peak than average and levels may remain elevated in the blood for longer.

Dr Collinson confirmed that when troponins started to replace the traditional cardiac enzymes in categorising cardiac events, that it was more likely that they would have been taken closer to when they would have reached peak values, when compared to practices nowadays. This was because they were more expensive and because they were not as sensitive when compared to the assays used today. Therefore, it would be more common practice to allow approximately 6 to 9 hours after the commencement of symptoms when troponin was measured in order to allow a greater time for the values to demonstrate an increase.

The introduction of the high sensitivity troponins enables accurate results to be acquired sooner, usually between 3 to 6 hours after the commencement of symptoms and allows the patient to be treated more promptly. However, this means that they are more often taken at a time much before the peak values would be obtained.

The extent of troponin release is also influenced by other mechanisms or factors, examples include but are by no means limited to; speed of intervention, degree of myocardial cell reperfusion, presence of collateral circulation, underlying etiology. An important consideration when reviewing myocardial infarction claims is to note when possible the time the troponin or other cardiac biomarker tests were performed and if these tests were attended prior or subsequent to any coronary interventions such as angiography, angioplasty or stenting. This is because coronary interventions may increase the level of troponin or enzymes which are detected.

Appendix 2 shows guidance for patients admitted with suspected cardiac ischaemia and is a good example of how the hospital concerned uses Troponin T in practice. It states the following:

- blood must be collected at 6 hours post admission or after the onset of chest pain to allow for the release and detection of Troponin.
- this particular example has a 99th percentile Troponin T assay at 6 hours of <14 ng/L.
- if Troponin T at 6 hours is ≥ 100 ng/L this is consistent with MI.
- if the 6 hour Troponin T result is ≥ 14 ng/L but <100 ng/L to repeat the test after 12 hours (provided there are other features of cardiac ischaemia)

The guidance says that as there is no consensus recommendations on the troponin T change between serial samples, it goes on to say that the National Institute for Clinical Excellence (NICE) suggest a 20% increase between samples to distinguish unstable angina from evolving MI. However, these guidelines are due to be updated and whilst they may have been applicable to previous troponin assays, it is likely that the change in serial readings will be greater, because of the increased sensitivity and lower figures relating to where the 99th percentile references that apply to hs troponins. Various increments of have been discussed ranging from 20% to 100% and above.⁵

The details provided in this example are similar to others used but there are variations in approach with regards to some aspects. For example, some hospitals suggest that troponin should be taken at presentation and at 6 hours and used an assumed myocardial infarction value of ≥ 50 ng/L rather than ≥ 100 ng/L.

Therefore, it can be seen that in cases where the troponin levels are significantly lower than our definition states, MI will be diagnosed and it very possible that there is no repeat taking of the troponins when they are more likely to reach their peak. Where no repeat troponin is taken, it presents a particular challenge for claims assessment.

Claims practices where troponin levels are below the claims criteria

Where troponin levels are recorded as being elevated but below the claims criteria thresholds, claims assessors will examine very closely all of the evidence such as symptoms, ECGs, angiography results etc., firstly to ensure that it is a case of acute coronary syndrome rather than the troponins being elevated for another reason (as is more likely at these lower levels) and then to estimate the likelihood that peak troponins would have met the values required in order to meet the definition.

The “Universal definition of myocardial infarction” paper referred to earlier, includes excellent detail with regards to physical symptoms and investigations that help to establish the diagnosis of myocardial infarction and is a document that provides one of the best sources for claims assessors of how findings should be considered.

It is also important that cases where the claims criteria are not fully met are referred to Chief Medical Officers and in particular if there is the availability of a cardiologist to review such cases.

The importance of ECG findings

The ECG findings are of particular importance in confirming the diagnosis of myocardial infarction, particularly when these findings are brought together with chest pain that is characteristic of ischaemia and can also provide some indication of the severity of the event.

Some consideration can be given where there is persistent ST segment elevation (>20 minutes) during an attack of acute coronary syndrome. This finding indicates that there is an acute blockage in an artery that requires urgent treatment with the intent of re-establishing the blood flow, usually by means of immediate angioplasty, referred to as “Primary” Percutaneous Coronary Intervention (PCI).

Where there is a diagnosis of ST segment Elevation Myocardial Infarction (STEMI), it is generally considered these events are of a severity where troponin levels required to meet claims criteria would be breached if taken at their peak. Although there is no research available that confirms this, it is the feeling within the specialists we have consulted that perhaps almost 100% of these cases would be found to meet the troponin levels within the current definition. It should also be remembered that with STEMI, troponin may not always be taken as already has been mentioned.

If there is no persistent ST segment elevation, then it is an indication that the artery is not completely occluded. However, they are still considered medical emergencies where the initial treatment is focused on stabilisation of symptoms. The ECG changes associated with evolving Non STEMI (NSTEMI) are often ST segment depression or T wave inversion. However, it is possible that NSTEMI may occur without any significant ECG changes whatsoever.

After acute myocardial infarction the heart muscle may recover and leave evidence of scarring on the heart muscle detectable on ECG that show the characteristic changes in the form of pathological Q waves. Again, this would indicate an event that is highly likely to be of a severity that would meet the troponin criteria if taken at peak. However, when considering Q waves, caution should be given to ensure that the event is likely to have occurred during the currency of the policy. With the majority of cases encountered at claim, this is usually very obvious.

Other ECG changes following myocardial infarction include T wave inversion and Left Bundle Branch Block. However, these are also found with other heart disorders and need careful evaluation. Provided there is other supportive evidence indicating recent acute myocardial infarction, then again these findings are significant in determining the severity of the event, where again it is likely troponin levels taken at peak would be satisfy current claims criteria.

In view of the importance and value that ECGs can provide, where appropriate serial ECGs throughout the length of hospital stay can provide excellent objective data in order to make an assessment in case where the other evidence is wanting. This is an option that claims

assessors should be aware of and consider as it can be far more helpful than just receiving a single ECG from one moment during an episode.

Coronary angiography

In cases where troponin elevation and other findings are not completely conclusive, details of angiography can be extremely useful as it will confirm the presence or absence of coronary artery disease and where present its severity. Where the findings do not detect any significant disease, then it substantially reduces the chances of myocardial infarction being a cause of elevated troponin, particularly when there is only mild elevation.

Imaging techniques

Non-invasive imaging techniques can also have a role in determining whether myocardial infarction has occurred particularly in the presence of elevated troponins. They can also sometimes be helpful in determining the size of events. However, they are not as able to show heart muscle damage of a very minor scale that currently only troponin is able to do.

Echocardiograms following myocardial infarction can be performed and may be useful in showing reduced ejection fraction (EF) that has significant importance with regards to prognosis. In addition, where findings show a definite reduction in expected ejection fraction following a MI, then it is indicative that a significant event has occurred that would very likely result in the criteria with regards to troponin levels being met if they were to be recorded at peak time.

Echocardiograms may also show heart wall motion abnormalities relating to coronary artery disease such as Left Ventricular (LV) dysfunction, usually described as areas showing hypokinesis, akinesis or dyskinesis. However, this information should be evaluated very carefully as echocardiography cannot distinguish whether these changes are due to myocardial ischaemia with or without MI. In addition, it should also be considered that findings may not be able to show if damage to the muscle is recent or otherwise.

Regional myocardial wall abnormalities may also be caused by non-ischaemic causes such as cardiomyopathy and inflammatory conditions, which is why the presence or otherwise of elevated troponins is important when considering this information.

Radionuclide imaging is more accurate than echocardiogram in determining where there are areas of the heart that have infarcted and where there is “reversible” ischaemia in order to determine treatment regimes. It is also useful in determining the amount of damage the heart muscle has suffered following MI.

Magnetic resonance imaging and X-Ray computed tomography can be of value in confirming myocardial infarction diagnosis. However, these are rarely performed. With the technology of these types of tests becoming increasingly sensitive, they have the potential that they could become more relevant in time.

Troponin elevation not associated with myocardial infarction

The Universal definition of myocardial infarction paper also details a number of conditions that might cause troponins to be elevated. They include pulmonary embolism, cardiac arrhythmias, myocarditis, aortic valve disease, renal failure and even extreme exertion.

Another useful reference that details potential causes of troponin increase is titled “*Troponin elevation in coronary vs. non-coronary disease*”⁵. This paper is interesting as it provides a good insight into the level of troponin increase that is likely with different events and shows

clearly that there are fewer alternatives to myocardial infarction when there is more significant cellular damage to the heart muscle.

Conclusion

We hope that the information provided here will help claims assessors to fully understand the issues surrounding the recent diagnostic developments related to MI and to provide guidance that can help to ensure consistency of approach where common definitions are being used, which is important from an industry perspective.

The ABI feels that at this stage there is no urgency to change the current heart attack definition contained within the current Statement of Best Practice for Critical Illness until the next review in 2014 for the following reasons:

- At this point the issues we have with the current heart attack definition are only affecting a very small number of claims
- To change the definition now would require full consultation with all interested parties and would be in addition to the full review of the Statement of Best Practice for Critical Illness in less than a year's time.
- There is time now to allow full consultation during 2013 with the full facts available before the 2014 review.
- Companies can still change definitions sooner if they wish to

Hopefully this paper addresses areas that need to be taken into consideration and will help companies to form a view ahead of the forthcoming 2014 review of the Statement of Best Practice for Critical Illness.

Lastly the group would like to sincerely thank Dr Paul Collinson for his considerable help in contributing towards this paper.

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Appendix – 1

Previous ABI model wordings for Heart Attack

ABI statement of Best Practice for Critical Illness Cover 1999:

Heart Attack

The death of a portion of the heart muscle as a result of inadequate blood supply as evidenced by an episode of typical chest pain, new electrocardiograph changes and by the elevation of cardiac enzymes.

The evidence must be consistent with the diagnosis of heart attack.

ABI Statement of Best Practice for Critical Illness in 2002:

Heart attack

The death of a portion of heart muscle, due to inadequate blood supply, that has resulted in all of the following evidence of acute myocardial infarction:

- typical chest pain;
- new characteristic electrocardiographic changes;
- the characteristic rise of cardiac enzymes, troponins or other biochemical markers;

where all of the above shows a definite acute myocardial infarction. Other acute coronary syndromes, including but not limited to angina, are not covered under this definition.

Appendix - 2

Troponin T

(Biochemistry)

Investigation : **Troponin T**
 Specimen type : **Serum**
 Spec container : **Serum Gel**
 Volume required : **5ml**

Reference range : **<14 ng/l**
 Turnaround : **<24 hours**

Knowledge of the time of on-set of chest pain is essential for interpretation and sampling. Samples must be taken at **LEAST** 6 hours post chest pain.

High Sensitivity Troponin T Assay

A new high sensitivity cardiac Troponin T assay is replacing the current Troponin T assay from Monday 25th October 2010. This assay is analytically more sensitive to lower levels of cardiac Troponin T than were measurable with the previous assay, it is also more precise with a 10% CV at <13 ng/L (less than the 99th percentile).

New Units

The high sensitive assay Troponin T assay will be reported in ng/L instead of ug/L (14 ng/L instead of 0.014 ug/L).

Diagnostic Criteria

The universal definition of myocardial infarction¹ depends on detecting an increase or decrease in Troponin T with at least one value above the 99th percentile of the upper reference limit, together with evidence of myocardial ischaemia. The 99th percentile using the new Troponin T assay is 14 ng/L.

Sample Timing

The new sensitive assay facilitates more rapid diagnosis of myocardial ischaemia. Blood must be collected at least 6 hours post admission or after the onset of chest pain to allow time for the release and detection of Troponin T in the circulation. Under certain circumstances it may be helpful to repeat the Troponin T in a further 6 hours.

Interpretation (additional information and procedural flowchart can be found [here](#).)

All Troponin T results must be interpreted in conjunction with clinical history and ECG findings.

99th percentile of Troponin T assay at 6 hours <14 ng/L

If the Troponin T at 6 hours is ≥100 ng/L this is consistent with myocardial infarction.

If the 6 hour Troponin T result is ≥14 ng/L but <100 ng/L or if there are other features of cardiac ischaemia, repeat the Troponin T at 12 hours post onset of chest pain or post admission (stating the timing on the request form) to assess if it is chronic or acute cardiac ischaemia.

There are currently no consensus recommendations on the Troponin T delta change between the serial samples, however current NICE guidelines suggest a 20% increase between the 1st and 2nd samples to distinguish unstable angina from evolving MI. This should however be interpreted with a full clinical history and ECG findings.

Mild Troponin T elevation without a significant change between results may be due to other non-ischaemic causes when there are no typical cardiac features. This is found in patients with renal failure, myocarditis or pulmonary embolus.

¹ESC/ACCF/AHA/WHF Task Force for the redefinition of Myocardial Infarction: universal definition of myocardial infarction. J Am Coll Cardiol 2007; 50:2173-95